Exercise courses

Start with Exercises 1 and 2. Then choose an exercise 3 or 4 without assistance.
to 7 are a reserve of transfer, research or programming: the professor designates one.
Exercise 8 resumes at a distance.

Exercise 9.1 · Transform

For all real x, we consider the expressions

$$A = e^{x}e^{2-x}, \qquad B = \frac{(e^{2x})^{3}}{e^{x}}, \qquad C = e^{-x}(e^{x}+1).$$

1. Simplify A and B justifying each grouping of exhibitors.

2. Develop and simplify C. Explain why a sum does not simplify in the same way a product.

Exercise 9.2 · Equations and inequities

An equation and a real unknown inequitable is considered x :

$$e^{2x-1} = e^{x+3} \quad \text{and} \quad e^{-3x} \geq e^{x-2}.$$

1. Resolve each in $\mathbb{R}$, citing the property of the exponential which allows comparison Exhibitors.

2. When solving the inequitable, explain at what stage the sense of inequality changes. all solutions in the form of interval.

Exercise 9.3 · Product and maximum

The function f is defined on [0; 4] by $f(x) = xe^{-x}$.

1. Calculate $f\prime(x)$, then study his sign over the interval.

2. Draw up the table of changes in f. Deduce its maximum and the abscisse where it is reached.

3. Check that the values at the bounds do not give a greater maximum.
exact values.

Exercise 9.4 · A modeled population

The following two parts are independent.

Part A · A population. A population is modelled by $P(t) = 200e^{0,1t}$where: $t \geq 0$ is the time in years. $P(t)$ is a theoretical workforce, which is not necessarily integer.

1. Determine initial strength, then multiplier coefficient between moments t and $t + 1$. In deduct the exact annual rate and its percentage rounded to one hundredth.

2. Explain why this model cannot necessarily describe a real population indefinitely.

3. **Part B · Two curves.** In the same coordinate system, outline the curves of $t \mapsto e^{\frac{t}{2}}$ and $t \mapsto e^{-\frac{t}{2}}$. Specify their variations, common point and symmetry. Clearly identify each curve.

Exercise 9.5 · Search for a threshold

The quantity of a substance, in milligrams, is modelled at the moment $t \geq 0$ per $Q(t) = 80e^{-\frac{t}{2}}$, where t is expressed in hours. Checks take place at full moments $n = 0, 1, 2, \ldots$. We're looking for the for which the quantity is strictly less than 10 mg.

1. Write a search algorithm of this first integer. Provide a bounder and specify what the program returns if the bound is reached.

2. Run the algorithm, then numerically check the amount in the row found and in the previous row.

3. To justify, with the help of the decrease of Q, that this double check establishes that it is indeed the do not use logarithm.

Exercise 9.6 · A symmetry

The function is considered f defined on $\mathbb{R}$ by $f(x) = e^x + e^{-x}$.

1. Studying gender parity f and interpret the result for its curve.

2. Demonstrate that $f(x) \geq 2$ for all real x, by choosing a method. Specify the case of equality.

3. Propose a second method of checking the result, from a square or from a study of changes.

Exercise 9.7 · Singleness under a hypothesis

Deepening. It is assumed that the exponential function has already been built. f a derivable function on $\mathbb{R}$ who verifies $f'(x) = f(x)$ for all real x and $f(0) = 1$. $g(x) = f(x)e^{-x}$.

1. Calculate g' and explain why g is constant on $\mathbb{R}$.

2. Determine this constant and then deduce from it the expression of f.

3. Does this demonstration establish the existence or uniqueness of the function sought? Justify.

Fiscal year 9.8 · Deferred review

The sequence is defined (u_n) by $u_n = e^{2n+1}$ for all natural n.

1. Demonstrate that this sequence is geometric, then specify its first term and common ratio.

2. Calculate $u_0 + u_1 + u_2$ In exact form. Compare the direct calculation and use of a formula of the sum.

Personal work. Perform only the task assigned at the end of the meeting, in progress
Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions
Resuming exercise 8 a week after the chapter (10 min).
is exceeded, note the difficulty and bring the tests; do not accumulate the whole reserve.

To be done in the CQFD manual

These exercises are dealt with in your manual. The progress indicates the session; they replace the personal work planned that day (10 to 20 minutes in total).

CQFD, exercise 18 p. 187, questions a, b and c. Simplify products and quotients by quoting the ownership of the exponentials employed.

CQFD, exercise 30 p. 187, questions a, b and c. Solve the three inequities and write the Sets of solutions in the form of intervals.